

Response Letter with Annotations

Zhao, J., Wei, Y., & Wu, B. (2022). Analysis of the social network and the evolution of the influence of ancient Chinese poets. *Social Science Computer Review*, 40(4), 1014–1034. <https://doi.org/10.1177/08944393211028182>

Annotations

1. Purposes

- a. Motivation (one-sentence summary): The article aims to build a large-scale social network of ancient Chinese poets and use computational network measures to quantify how poets' influence evolved across dynasties.
- b. Specific Rationale (several bullet points):
 - Prior work in digital humanities had usually focused on a single poet, a single dynasty, or one isolated aspect of literary history rather than a long-run, cross-dynasty picture.
 - Ancient poetry is deeply social: poets wrote to friends, admired predecessors, referenced one another, and developed schools/styles through interaction.
 - Because poet relationships are overlapping, large in scale, and distributed across poems and biographies, the authors argue that computational social network analysis is needed.
 - Name ambiguity and alias use are a major obstacle in classical Chinese texts, so alias matching is treated as a prerequisite for building the network.
 - The authors also want to move beyond static snapshots and model influence as something that changes over time.
- c. Research Questions and/or Hypotheses: The article does not state formal research questions or hypotheses. Instead, it is framed as an **exploratory** computational study to explore poets' social networks and the evolution of their influence across dynasties using complex network analysis and entropy-based weighting methods.
 - Identify IV, DV, mediator, moderator
 1. IV: degree centrality, closeness centrality, betweenness centrality, eigenvector centrality, and number of poems
 2. DV: poet influence score within each dynasty/time period
 3. Beside IV and DV, the verification experiments treat SI-model diffusion performance as an external criterion for judging whether a ranking method identifies structurally important nodes.
 - Identify specific relationships among variables
 1. Higher values on the five node/network indicators contribute to a higher composite influence score after entropy-based weighting
 2. TSEWM adds a time-series adjustment so that indicator weights in period t are informed by weights in $t-1$, producing a smoother temporal evolution curve

3. In the verification stage, higher-ranked nodes are expected to generate faster and broader SI diffusion if the ranking method is effective.

2. Method

a. Research Design

- Social network analysis
- Verification based on network diffusion simulations
- Time is operationalized at the dynasty level (and, for some analyses, subperiods within the Tang and Song dynasties).

b. Sample

- Poetry data: 905,675 works by 41,310 poets from the Pre-Qin period to the Qing dynasty.
- Network size: 41,310 nodes and 114,697 edges in the full network; the maximum connected subgraph contains 40,359 nodes and 114,156 edges.
- Sources: Souyun, gushiwen.org, Wikipedia, Dictionary of Chinese People's Names in Past Dynasties, comments/collections on Tang and Song poetry, and CBDB

c. Measures

- Node = poet. Edge = social relation or citation/reference relation between poets
- Biographical relations: extracted via named entity recognition and relation extraction from poet biographies; example relation types include kinship, friendship, love, teacher–student, admiration, and partner.
- Alias matching: used CBDB plus Wikipedia-based extraction for periods not covered by CBDB; additional cleaning removed ambiguous or overly common aliases
- Weighted edges: based on the number of times one poet referenced another in titles or poem bodies (multiple mentions within one poem counted once).
- Influence indicators (Table 2): degree centrality, closeness centrality, betweenness centrality, eigenvector centrality, and number of poems.
- TSEWM procedure: min – max standardization → entropy-based weighting of indicators within each dynasty → time-series adjustment of weights → weighted sum to compute influence score

d. Analysis

- Network visualization in Gephi and descriptive network statistics (e.g., sparsity, average degree, clustering coefficient, average shortest path length).
- Ranking of poet influence across dynasties and within selected schools/periods.
- Temporal evolution analysis for Confucius and for Tang/Song poets or schools.

- Verification Experiment 1: compare EWM with degree, PageRank, and k-shell using SI diffusion curves seeded by the top 1% of nodes.
- Verification Experiment 2: correlate node ranking with number of infected nodes after $t = 5$ SI iterations.
- Verification Experiment 3: compare ordinary EWM vs. TSEWM using Confucius' influence trajectory as an ablation-style demonstration.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- The study successfully constructs a very large, cross-dynasty poet social network, which the authors present as the most complete such network spanning more than 2,000 years.
- The maximum connected subgraph covers 99.7% of nodes, and the network shows small-world characteristics (average shortest path length = 6.454 in the maximum connected subgraph).
- Tang, Song, Ming, and Qing dynasties together account for more than 80% of the network; Tang poets occupy a particularly dense portion of the graph.
- In the influence ranking, Tang and Song poets are generally at a high level overall; the authors highlight cases where contemporaneous influence may differ from later canonical fame.
- Confucius' influence is shown as lowest in Qin, then rising and peaking in Song; Tang influence is split into early/prosperous/middle/late stages; Song schools are compared as graceful/restrained vs. bold/unconstrained.
- Across the verification experiments, EWM/TSEWM is reported to outperform degree, PageRank, and k-shell in propagation speed, propagation range, and rank–diffusion correspondence.

b. Include Key Figures/Tables

- Table 3: Nodes and edges by dynasty.

Table 3. Nodes and Edges Information of Poet Social Network.

Dynasty	Nodes	Edges (Same Dynasty)	Edges (Different Dynasty)	Edges
XianQin	210	208	81	290
Qin	43	42	101	143
Han	615	690	352	1,042
WeiJin	505	594	743	1,337
Sui	240	231	451	682
Tang	4,952	11,629	2,796	14,425
NanBei	961	1,351	470	1,821
Song	8,958	25,004	4,499	29,503
Liao	26	25	8	33
Jin	615	262	353	705
Yuan	1,707	2,855	2,208	5,063
Ming	7,446	27,060	7,025	34,085
Qing	11,253	23,508	8,135	31,643
Sum	37,531	93,459	27,222	120,772

- Table 4: Overall network parameters

Table 4. Topological Parameter of the Poets' Social Network.

Type of Network	Number of Nodes	Number of Edges	Sparsity	Average Degree	Average Clustering Coefficient	Average Shortest Path Length
Network	41,310	114,697	.000134	5.553	.057	—
Maximum connected subgraph	40,359	114,156	.00014	5.657	.058	6.454

- Figure 3: Visualization of the poet social network across dynasties; node size reflects influence and edge weight reflects closeness of communication

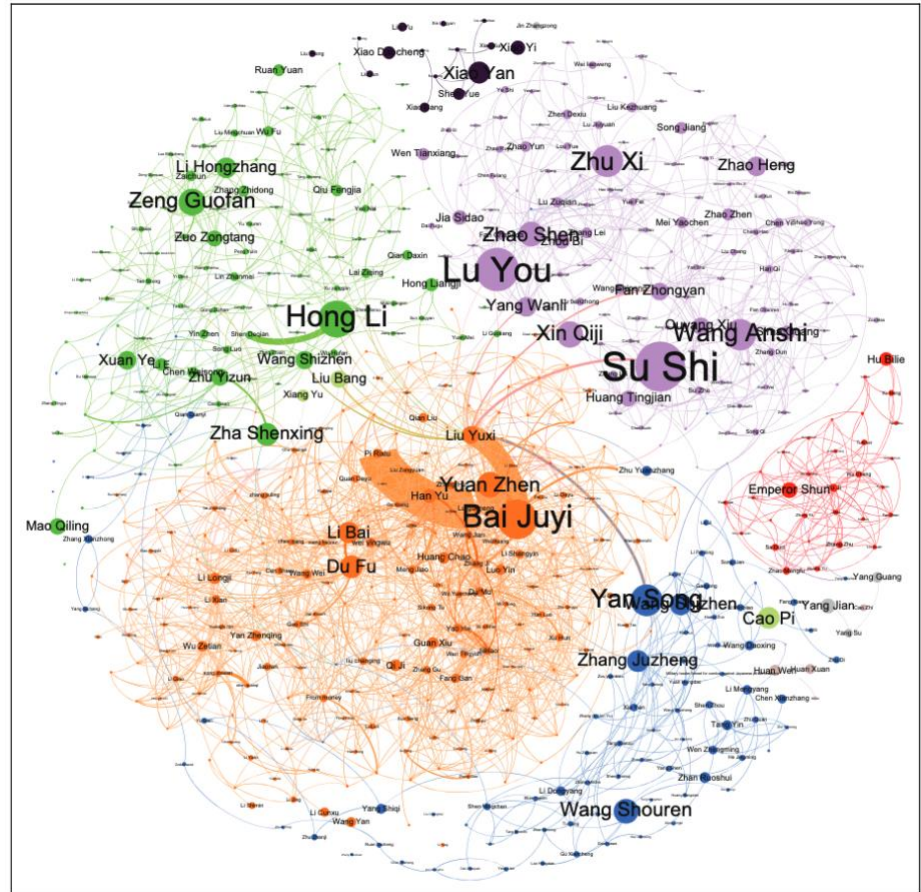


Figure 3. The social network of poets. Note. The different colors of the nodes in the picture represent the different dynasties of the poets (orange: Tang Dynasty; purple: Song Dynasty; blue: Ming Dynasty; green: Qing Dynasty), the size of the node indicates the poets' influence, and the weight of the edges indicates how closely the two poets communicated with each other.

- Figure 4: Top five influential poets by dynasty

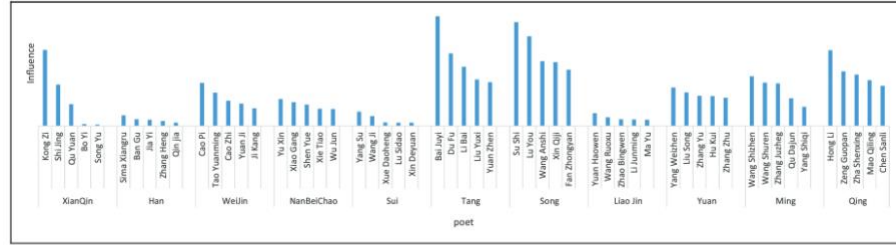


Figure 4. Ranking of poets' influence in different dynasties. *Note.* The top five influential poets of different dynasties are selected. The abscissa is the poet's name, which is sorted by dynasty time from left to right, and the ordinate represents the poets' influence. Because of the short period of Qin Dynasty and the backward development of literature, it is not shown here.

- Figures 10: SI-based verification plots showing EWM's stronger diffusion performance and clearer negative ranking–diffusion relationship

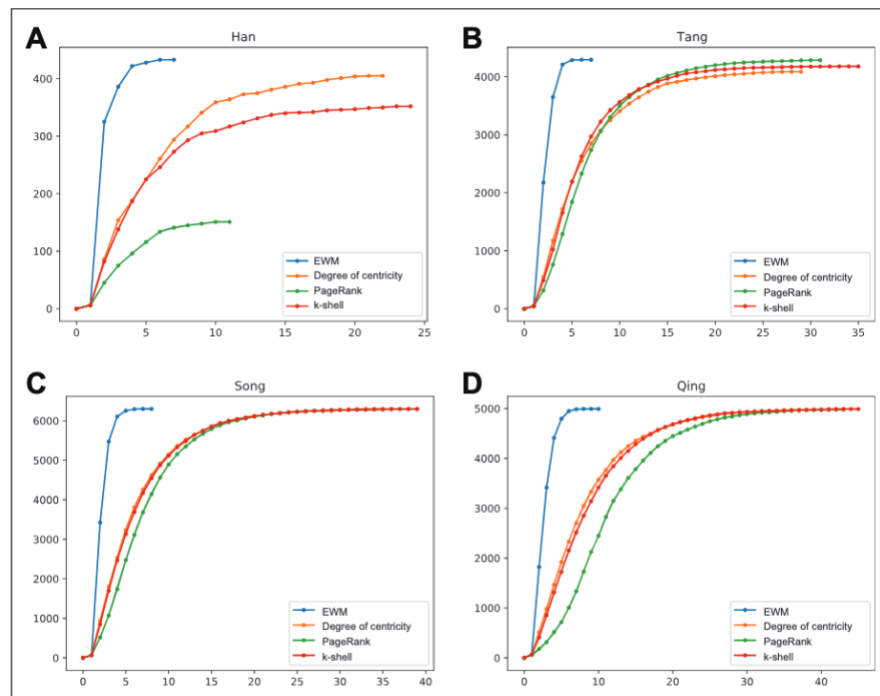


Figure 10. Susceptible–infected transmission curve when the top 1% nodes obtained by each measurement method are the nodes of the initial source of infection. *Note.* The abscissa represents the number of time iteration steps, and the ordinate represents the number of infected nodes when reaching steady state, which illustrates the propagation range of infected nodes. Panel A: Han Dynasty. Panel B: Tang Dynasty. Panel C: Song Dynasty. Panel D: Qing Dynasty.

- Figure 15: Comparison of EWM and TSEWM, with the time-series method producing a smoother influence trajectory for Confucius

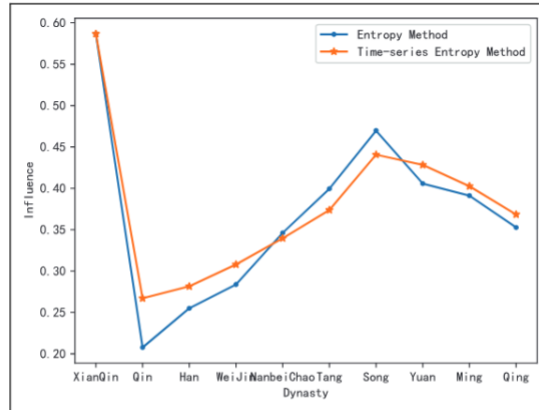


Figure 15. Comparison chart of influence evolution calculation based on entropy weight method (EWM) and time-series entropy weight method. Note. The blue line in the figure represents the evolution process of Confucius' influence calculated by the unmodified EWM, and the orange line represents the evolution process of Confucius' influence calculated by the EWM with time series.

c. Authors' Conclusions

- Computational network analysis can reveal long-run patterns in the development of ancient Chinese literature that are difficult to detect through manual reading alone.
- The TSEWM approach is presented as a more robust and temporally sensitive way to measure changing poet influence.
- The authors argue that modeling poet influence helps illuminate not only individual prestige but also the evolution of poetry schools, intellectual change, and aspects of ancient social life.

4. Discussion

a. Implications

- Demonstrates how multisource textual data and network methods can be used to build large-scale literary knowledge infrastructures.
- Offers a long-run, cross-dynasty view of poetic relations instead of isolating one poet or one era.
- Shows that canon formation and contemporaneous network prominence may not always align.
- Suggests that literary history can be studied as an evolving relational system rather than only a sequence of individual authors.

b. Limitations

- No spatial analysis: The study only examines temporal evolution across dynasties and does not incorporate geographic information or spatial diffusion of literary influence.
- Focus on individual influence only: The analysis evaluates influence at the individual node level and does not investigate group-level dynamics, literary communities, or collective influence mechanisms.
- Limited data sources: The poet relationship data mainly comes from several historical texts, which may lead to incomplete or biased network construction

c. Future Directions

- Expand data sources and include spatial information (e.g., locations from biographies and local chronicles).
- Study group-level influence among poets and build a time–space network of ancient literature

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Response Letter (~600 words)

- Your assessment of the study

This article offers an ambitious and creative attempt to apply computational methods to the study of ancient Chinese literature. By integrating multisource data, the authors construct a large-scale social network of 41,310 poets spanning more than 2,000 years and use network analysis to examine poet relationships and the evolution of influence across dynasties. The study is particularly valuable for showing how digital humanities methods can extend conventional literary research beyond close reading or single-dynasty case studies. Instead of focusing only on one poet, one period, or one literary school, the article seeks to provide a macro-level and long-term view of literary development. In this sense, the study is impressive in scope and demonstrates the promise of combining natural language processing, social network analysis, and historical text mining to study literary history. The visualizations and dynasty-based comparisons are also useful in making broad historical patterns more visible and interpretable.

Another strength of the article is that it attempts not only to construct a network, but also to quantify poet “influence” over time. The proposed time-series entropy weight method (TSEWM) reflects the authors’ effort to move beyond static network analysis and capture temporal evolution, which is an important and interesting goal for literary history. The article therefore contributes methodologically by introducing a computational framework that future scholars could refine, adapt, or challenge.

- Weaknesses

At the same time, the article has several important weaknesses. First, the concept of “influence” is not fully aligned with how it is operationalized. The authors cite Rashotte’s sociological definition of influence as a process in which individuals change others’ thoughts, feelings, attitudes, or behaviors through interaction. However, the empirical analysis does not measure such change directly. Instead, “influence” is approximated through centrality measures, poem counts, and entropy-based weighting. As a result, the study appears to measure structural prominence or network importance rather than social influence in the stronger theoretical sense introduced earlier. This creates a noticeable gap between the conceptual definition and the actual computational operationalization.

Second, the network construction strategy is somewhat crude. A relationship is inferred whenever one poet’s name appears in another poet’s biography or poetic text, and then different ties are merged into an undirected network. While this makes large-scale modeling possible, it may oversimplify literary relationships. Mentioning another poet does not always indicate meaningful social connection, and different relationships such as admiration, friendship, citation, kinship, and teacher–student ties may have very different meanings. Converting all of these into largely comparable edges risks flattening important distinctions in literary history.

Third, the validation strategy is not entirely convincing. The SI propagation experiments are presented as evidence that EWM or TSEWM better captures node importance, but these experiments mainly show that nodes selected by the proposed method perform better under a specific diffusion model. This does not necessarily prove that the method captures literary influence more accurately. In addition, the paper does not clearly report some simulation details, such as parameter choices and robustness checks, which limits reproducibility. Similarly, the comparison between EWM and TSEWM relies heavily on the claim that a smoother influence

curve is preferable, but smoother curves are not automatically more valid historically. Abrupt changes in literary influence may sometimes reflect real historical dynamics rather than noise.

- Other notes

One useful aspect of the article is that the authors implicitly acknowledge some limitations through their future work section. They note that future research should expand the dataset, integrate location information from biographies and local chronicles, and move from individual-level influence to group-level influence mechanisms. This is important, because the current article does not yet include a true spatial dimension, even though its long-term goal is to build a complex time–space network of ancient literature. Future research that combines social ties, literary communities, and geographic movement would likely make the framework much stronger. Overall, this study offers a very valuable exploratory contribution: it is innovative and thought-provoking, although its theoretical framing and measurement choices can still be improved.